Power Supplies & Regulators

Exercise Book

IESE Exercise Book 3 of 3

Problems only — attempt them first; answers (final results, no worked steps) are at the back.

How to use this book

This is the problem set distilled from the *Power Supplies & Regulators* solving manual. It contains **only exercises and their answers** — no theory and no worked derivations. Each chapter mirrors a chapter of the manual, so when you are stuck on a *method*, open the manual at the same chapter and read its Protocol and Worked Example.

Key Idea

Sizing exercises run *backwards*, from the load to the wall: fix the regulator's headroom first, then the ripple, then the diode drops, then the worst-case mains. The answer key gives the standard part/value reached at the end — get there yourself before you look.

Contents

Chapter 1

Specifications: reading the data sheet

Exercise 1.1 A datasheet line by line

A $5\,\mathrm{V}$ regulator's table reads: $\Delta V_{out} = 10\,\mathrm{mV}$ for I_{out} from $5\,\mathrm{mA}$ to $1.5\,\mathrm{A}$; $\Delta V_{out} = 3\,\mathrm{mV}$ for V_{in} from $8\,\mathrm{V}$ to $12\,\mathrm{V}$. (a) Load and line sensitivity, and the ripple rejection in dB. (b) At $V_{in} = 12\,\mathrm{V}$, $I_{out} = 1\,\mathrm{A}$: efficiency and dissipated power.

Exercise 1.2 Reading the numbers

A regulator shows $\Delta V_{out} = 7\,\mathrm{mV}$ over an input range $\Delta V_{in} = 17.5\,\mathrm{V}$, and $\Delta V_{out} = 30\,\mathrm{mV}$ for a load step $\Delta I_{out} = 1.5\,\mathrm{A}$. Find the ripple rejection (dB) and the output resistance.

Chapter 2

The classical (transformer) power supply

Exercise 2.1 Size the secondary

Size the supply for $12\,\mathrm{V}/0.5\,\mathrm{A}$ from a 7812 ($V_{DO,min} = 2\,\mathrm{V}$), full-wave bridge with $V_D = 1\,\mathrm{V}$, ripple $\Delta V \leq 1\,\mathrm{V}$, mains $230 \pm 10\,\%$ at $50\,\mathrm{Hz}$. (a) The filter capacitor. (b) The secondary RMS voltage and the turns ratio. (c) Why would a *half-wave* rectifier double the capacitor?

Exercise 2.2 Filter capacitor

Full-wave, $f = 50\,\mathrm{Hz}$, $V_{MAX} = 9\,\mathrm{V}$, $I_{out} = 100\,\mathrm{mA}$, $C = 2.2\,\mathrm{mF}$ (use $T_{dis} = T$). Find the ripple ΔV, the conduction time t_c and the peak charging current I_{peak}.

Exercise 2.3 The capacitor trade-off

Repeat the previous exercise with $C' = 4.7\,\mathrm{mF}$ (same f, V_{MAX}, I_{out}). What happens to ΔV and to I_{peak}?

Chapter 3

Power factor and its correction

Exercise 3.1 R–L load and its correction

$R = 3\,\Omega$, $L = 10.6\,\mathrm{mH}$, $V_S = 480\,\mathrm{V_{RMS}}$ at $f = 60\,\mathrm{Hz}$ ($\omega = 377\,\mathrm{rad/s}$). (a) Find $|I_S|$, φ, S, P and the power factor. (b) Find the parallel capacitor that corrects PF to 1, and the new source current.

Exercise 3.2 The pure inductor

A purely inductive load $L = 10.6\,\mathrm{mH}$ (no R), same source as above. (a) Find $|I_S|$ and the power factor. (b) Find the parallel C that brings the source current to zero.

Chapter 4

Linear voltage regulators

Exercise 4.1 7805 at light load

A 7805 supplies $I_{out} = 200\,\text{mA}$ from $V_{in} = 8\,\text{V}$. (a) Dissipation, efficiency, and is the bare package ($R_{th} = 60\,\text{K/W}$) enough at $T_{amb} = 25\,°\text{C}$? (b) What is the lowest rippled-input *valley* that still regulates?

Exercise 4.2 7805 needs a heat sink

A 7805 supplies $I_{out} = 1\,\text{A}$ from $V_{in} = 12\,\text{V}$. Find the dissipation and efficiency, and the junction temperature rise both bare ($60\,\text{K/W}$) and with a $10\,\text{K/W}$ heat sink.

Exercise 4.3 Efficiency ceiling

For a 78xx ($V_{DO,min} = 2\,\text{V}$), give the maximum efficiency $\eta_{MAX} = V_{out}/(V_{out} + 2)$ for $V_{out} = 3.3\,\text{V}$, $5\,\text{V}$ and $12\,\text{V}$.

Chapter 5

Switching regulators

Exercise 5.1 Buck duty cycle

A buck converter makes $V_{out} = 5\,\mathrm{V}$ from $V_{in} = (9 \pm 1)\,\mathrm{V}$. Give the nominal duty cycle δ and its range over the input range.

Exercise 5.2 Boost from a battery

A $12\,\mathrm{V}$ rail must come from a battery $V_{in} = (5 \pm 0.5)\,\mathrm{V}$, $f_{SW} = 200\,\mathrm{kHz}$. (a) Topology and the δ range. (b) With a capacitor-current swing $\Delta I_C = 1.2\,\mathrm{A}$ and $\mathrm{ESR} = 50\,\mathrm{m}\Omega$, the output ripple. (c) Why is this topology's feedback loop deliberately made *slow*?

Exercise 5.3 Minimum L for CCM

A buck has $V_{in} = 9\,\mathrm{V}$, $V_{out} = 5\,\mathrm{V}$ ($\delta = 0.556$), $f_{SW} = 100\,\mathrm{kHz}$, lightest load $I_{out,min} = 50\,\mathrm{mA}$. Find the smallest L that keeps it in CCM and give the next standard value.

Chapter 6

Laboratory: Buck converter

Exercise 6.1 CCM or DCM?

For a buck with the boundary (critical) current $I_B = \dfrac{V_{out}(V_{in} - V_{out})}{2Lf_{sw}V_{in}}$, decide the conduction mode: (a) $V_{in} = 8\,\mathrm{V}$, $V_{out} = 5\,\mathrm{V}$, $R_L = 30.5\,\Omega$, $I_B \approx 0.11\,\mathrm{A}$; (b) $V_{in} = 24\,\mathrm{V}$, $V_{out} = 5\,\mathrm{V}$, $R_L = 50\,\Omega$, $I_B \approx 0.23\,\mathrm{A}$.

Chapter 7

Answers

Final results only — the matching chapter of the *manual* shows how each is obtained.

Chapter 1 — Specifications

1.1 (a) $S_I = 0.010/1.495 \approx 6.7\,\text{m}\Omega$; $S_V = 0.003/4 = 0.75\,\text{mV/V}$; ripple rejection $= 20\log_{10}(4/0.003) = 62.5\,\text{dB}$. (b) $P_{out} = 5\,\text{W}$, $P_{in} \approx 12\,\text{W}$: $\eta = 41.7\,\%$, $P_d = 7\,\text{W}$ of heat.

1.2 Ripple rejection $= 20\log_{10}(17.5/0.007) = 68\,\text{dB}$; $S_I = 0.030/1.5 = 20\,\text{m}\Omega$.

Chapter 2 — The classical (transformer) power supply

2.1 (a) $C = I_{out}(T/2)/\Delta V = 5\,\text{mF}$. (b) $V_{MAX} = 15\,\text{V}$, secondary peak $17\,\text{V}$, $V_{S,RMS} = 12.0\,\text{V}$; at $-10\,\%$ that is $13.4\,\text{V}$ $\Rightarrow$ pick a $15\,\text{V}_{\text{RMS}}$ secondary, ratio $\approx 230/15 = 15.3$. (c) Half-wave discharges for the whole period T, twice as long, so the same ΔV needs $C = 10\,\text{mF}$.

2.2 $\Delta V = 0.9\,\text{V}$, $t_c = 1.44\,\text{ms}$, $I_{peak} = 2.77\,\text{A}$.

2.3 $\Delta V' = 0.43\,\text{V}$ (smaller), $t'_c = 0.98\,\text{ms}$, $I'_{peak} = 4.1\,\text{A}$ (larger) — lower ripple, higher current peak.

Chapter 3 — Power factor and its correction

3.1 (a) $\omega L = 4\,\Omega$, $|Z| = 5\,\Omega$, $|I_S| = 96\,\text{A}$, $\varphi = 53.1°$, $S = 46.08\,\text{kVA}$, $P = 27.67\,\text{kW}$, $PF = 0.6$. (b) $C = L/(\omega^2 L^2 + R^2) = 424.5\,\mu\text{F}$; source current drops to $58\,\text{A}$, $PF = 1$.

3.2 (a) $|I_S| = V_S/(\omega L) = 120\,\text{A}$, $\varphi = 90°$, $PF = 0$. (b) $C = 1/(\omega^2 L) = 663.8\,\mu\text{F}$ (resonance), ideally $I_S = 0$.

Chapter 4 — Linear voltage regulators

4.1 (a) $P_{reg} = 0.6\,\text{W}$, $\eta = 62.5\,\%$; junction rise $36\,\text{K}$ $\Rightarrow T_j \approx 61\,°\text{C}$ — fine without a heat sink. (b) $V_{MIN} = V_{out} + V_{DO,min} = 7\,\text{V}$.

4.2 $P_{reg} = 7\,\text{W}$, $\eta = 42\,\%$; bare rise $420\,\text{K}$ (far past the $125\,°\text{C}$ limit), with $10\,\text{K/W}$ rise $70\,\text{K}$ $\Rightarrow T_j \approx 95\,°\text{C}$ (acceptable).

4.3 $\eta_{MAX} = 62.3\,\%$ (3.3 V), $71.4\,\%$ (5 V), $85.7\,\%$ (12 V).

Chapter 5 — Switching regulators

5.1 $\delta = V_{out}/V_{in} = 55.5\,\%$; range $50\,\%$ ($V_{in} = 10\,\mathrm{V}$) to $62.5\,\%$ ($V_{in} = 8\,\mathrm{V}$).

5.2 (a) $V_{out} > V_{in}$: **Boost**, $\delta = 1 - V_{in}/V_{out}$, range $62.5\,\% \ldots 54.2\,\%$ (nominal $58.3\,\%$). (b) $\Delta V_{out} = \Delta I_C \cdot \mathrm{ESR} = 60\,\mathrm{mV}$. (c) The boost is prone to instability, so its loop is given a low bandwidth — it reacts slowly but will not oscillate.

5.3 $L_{min} = \dfrac{(V_{in} - V_{out})\delta}{2 I_{out,min} f_{SW}} = 222\,\mu\mathrm{H}$; pick the next standard $330\,\mu\mathrm{H}$ ($\Delta I_L \approx 67\,\mathrm{mA}$, CCM holds).

Chapter 6 — Laboratory: Buck converter

6.1 (a) $I_{out} = V_{out}/R_L \approx 0.16\,\mathrm{A} > I_B \Rightarrow$ **CCM**. (b) $I_{out} \approx 0.10\,\mathrm{A} < I_B \Rightarrow$ **DCM** (light load at a high step-down ratio).